A comparison of thoughts on the Russo-Ukrainian war 2 years apart using topic modeling and sentiment analysis

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Abstract

This paper studies the online discussion of the Russo-Ukrainian war on Reddit, using the natural language processing module in python, NLTK, to conduct sentiment analysis using its sub-module VADER. This study finds that the sentiment towards Russia has more or less stayed unchanged while the sentiment towards Ukraine has grown in positivity. Using another NLP module in python, BERTopic, topic modeling on the data was carried out, finding that although many of the topics overlap, the popularity of these topics has changed. There has been a shift in the perspective and emphasis of discussion on the attacks and conflicts of the war, going away from the more political thoughts on the topics beforehand.

1 Introduction

Since 24th February 2022, Ukraine and Russia have been engaged in an armed conflict after Russia invaded Ukraine on three fronts with 190,000 troops, moving under the premise that the regions of Donetsk and Luhansk were "independent people's republics" [Ell24]. Due to the nature of the conflict involving one of the world's superpowers against a smaller country and the aggressive nature of Russia's invasion, public opinion has largely remained "pro-Ukraine" and "anti-Russia". This study aims to check if the opinions of users have gotten further negative compared to before considering the fact that the war has been going on for a few years with Ukraine often remaining on the defensive. Many major events such as the Ukraine counteroffensive in 2022, June 2023, and the growing cybersecurity war between the 2 countries. [Suf23]. We aim to study the discussions of users on various Reddit communities about the Russia/Ukraine conflict. We aim to use the publicly available dataset on tconflict and also utilise newer text data mining [GK23] on Reddit to compare the two. The data collected was downloaded from several monthly Reddit submission and comment dumps on academictorrents from the months March to June [Wat, Rai24, Raia, Raib].

This paper aims to offer a new perspective on the opinions of people regarding the war first via the recency of the data that is going to be used (March-June 2024) especially because the collection of data around this topic has been quite sparse in the past few years, and perhaps as a result, the sentiment analysis, computational treatment of opinions, [MHK14] on this topic has also been staggered. There is a large amount of significance in keeping up with public opinion which changes at every moment due to events happening, posts getting recognition, or even misinformation tricking people. Understanding the opinions of people regarding this war is crucial for decision making in governments and other agencies involved in the war effort, allowing a deeper reading to be made about the war. Lastly, the attention on the war in the media has been falling despite Russia still having occupied Crimea and Ukraine still having major counter-offensive operations. Consequently, I wish for this paper to reignite the academic conversation regarding the war so that more attention can be brought back to this issue, potentially alleviating the lives of those affected.

Across social media, Russia has often been represented as the aggressor over the last couple of years, likely due to them orchestrating an invasion and resorting to war first. As a result, I have hypothesized that the sentiment towards Russia has only gotten more negative than the sentiment when the war first broke out, a time with a substantial amount of confusion and misinformation in the media. [BT22] Furthermore, I hypothesise that the popularity of the war has been curbed significantly

since the year it started. By significantly, I mean in specific that less than 50% of the previous volume of users are active now.

Hypothesis: The hypothesis is that users are more biased against Russia as it is represented as the aggressor on various press media over the past 2 years but that there are far less users engaged in discourse on the war.

Research Questions: We formulate the following two research questions towards addressing our hypothesis.

1. How does sentiment change over time towards Ukraine and towards Russia?
2. What topics do users discuss and how have they changed over time?

Contributions

I firstly use a newly collected dataset from the months of March to June of 2024, and then extract the required data, cleaning it using standard natural language processing techniques. Then I use VADER to analyse and compare the sentiment of Reddit users towards Ukraine and Russia 2 years apart using a dataset made by Zhu et al [ZHL+22], displaying how opinions may have changed over time. It also uses topic modeling via BERTopic to offer an overview on the types of discussions that happen in the subreddits for both the collected dataset and the researchers' datasets, focusing on how the emphasis of certain topics has shifted as the war has continued.

2 Background

We now provide a brief history of the Russia/Ukraine conflict and then we disucss the Reddit communities that we focus on in this study.

2.1 Russia/Ukraine conflict

Although the Russo-Ukrainian war officially started on the 24th of February 2022 after the invasion by Russia, major tension had been built since the Maidan uprising in 2013, when people of Ukraine carried out peaceful protests against the pro-Russian government [Ish20] that wanted to silence any form of protest through laws. Quickly, these protests turned violent with police firing upon hundreds of Ukrainians in protest. As a result of the government's actions, specifically Yanukovych's, the European Union intervened to mediate the situation in Ukraine, creating an interim government that promptly charged Yanukovych with mass murder, thus removing the pro-Russian government. [Ste24]

As a result, Russia began its retaliation which was in the form of an invasion of Crimea whose government was under Russian control by the morning of the 27th of February 2014. [Beb15] Since then, Putin has further been on an agenda to, in his words, restore the "Russian World" [S. 22] with the war being one of the more recent steps. So far in this war, due to the damage caused to buildings and the aid required by the civilians, the economic toll is estimated to be $638B in total by the World bank and the UN [Erg24].

There are still many attacks occurring on both Russia and Ukraine even at the time of this paper being written such drone strikes on Russia [Won24] or substation strikes from Russia, leaving Pokrovsk, a city, without power and with a precarious water supply. [Abd24] Evidently, the war is still in force and should be treated as seriously as it was at the beginning.

2.2 Reddit Communities

Reddit is a social media platform that forms a community where you can make and join sub-communities or "sub-reddits" which are collections of users that post/share data that relate to the title of the sub-reddit: for example news about discoveries on astrophysics on the official science sub-reddit. Each post can be "upvoted" or "downvoted" by users depending on their opinions or readings towards it. Users discover posts via the feed they receive on the home page when they open the application or website, allowing them to scroll and view thousands of posts. Each post has a title along with content which can include text or images. Furthermore, the communities are open and anyone can join in on the conversation, meaning that these communities are not echo chambers and therefore can offer a diverse range of opinions, improving the dataset and thus any results using it.

The most significant reason for choosing reddit data is the amount of information that can be extracted from each post. This allows for much deeper analysis on the opinions of people regarding the war. In each post the information is compiled into a json file which contains x number of values, allowing for many parameters to be tested/plotted including the number of upvotes a post has, the number of downvotes it has, or even the number of awards a post has. Therefore, the many underlying thoughts and ideas of users can be investigated and evaluated to a highly accurate degree. Furthermore, the character limit on a post is far smaller than other social media platforms, allowing for users to fully express anything they have to say within a single post, which is very valuable for data analysis. The subreddits/communites that we worked on was less of a choice but more of a necessity arising from the paper by Zhu et al that published the reddit dataset for this topic using the same ones; thus, we can compare and contrast the 2 sets of data. [ZHL+22] The data from this dataset was collected from February 24th 2022 - when the invasion started - to 13th June 2022, around 2 years before the span of my data.

Focusing on sentiment analysis was of importance because on social media sites like Reddit, people share opinions and thoughts not only very openly but also very often. Sentiment analysis perfectly takes advantage of this openness as people express their emotions far stronger, making data collection much more valuable. Furthermore, due to the war being sidelined in the media in favour of more topical events and headlines, especially with regards to the conflict between Israel and Palestine, I believe it is of equal importance that attention is not shifted from another front of war.

3 Literature Review

In this section, we discuss the existing work on conflicts around the world, but more specifically on the Russia-Ukraine conflict, before discussing works that perform topic and sentiment analyses on data similar to ours.

3.1 Conflict Discussion on Online Communities

In this age, the world, although mostly free from conflict, still has a number of wars, struggles, and confrontations that are active or even ones that are over yet have had a lasting effect on the locations of those issues. As a result, there are a number of studies focusing on the online discussion of various communities on these topics.

Another major conflict at the moment is the one between Palestine and Israel. This conflict has roots in 1948: a day after Israel's establishment, on May 15th 1948, 5 Arab states attacked Israel who then had to displace 700,000 Palestians from their homes. [Reu24] Since then a series of wars have been fought, increasing the Tensions between the 2 countries.

The communities engaging in conversation about the conflict between Palestine and Israel were

studied in a paper by Imtiaz A et al [IKLL22] where they aimed to study the stance of users regarding Israel and Palestine. This study, similar to mine, goes a step further in sentiment analysis and uses the VADER package from NLTK to detect stance of various social media posts, which in their case was used twitter. They found that majority of the posts in the USA were actually pro-Palestine and that many posts showed discontent to the USA government as they had reached a $735M deal of weapon sales to Israel. [Ale21]

Another conflict that recently ended was the Colombian conflict which was a civil conflict arising from the government unable to protect and help its people. Consequently, the Revolutionary Armed Forces of Colombia (FARC-EP) rose, rebelling against the government, its laws, and resulting in a conflict motivated by power [VSL18]. Due to this conflict, in the the period 1980-2000 there were around 25,000 deaths a year with an estimated 18% from political violence [LeG03].

In a study by Vera et al [VSL18], the post conflict situation in Colombia discourse online was studied using twitter data. The dataset used was very small, being 500 tweets with the already limited number of characters on each post so conclusions may be more biased and therefore less representative of the thoughts of the population. Their methodology was quite different to mine, most notably the fact that they did not use any python packages for sentiment analysis and NLP. Alternatively, they opted to use Text2data, an excel tool that does sentiment analysis, using pre-trained models. They generated word clouds on the foreigner tweets and the Colombian tweets separately, allowing for a general glimpse into the ideas discussed in the tweets.

3.2 Topic and Sentiment Detection

Text analysis is a sub-division of text mining: a process which involves extracting information from various written sources, in particular forming hypotheses by linking various findings together. [Hea03] Text analysis involves the actual derivation of high quality information from sources of texts [GCP14] and sentiment analysis is a type of text analysis that aims to extract information to do with opinions, preferences, or feelings expressed in text. There are many types of algorithms that attempt to do this such as a semantic approach [ZLG+11], the random walk algorithm [TW11], but the one that this study focuses on is the one that VADER - Valence Aware Dictionary for sEntiment Reasoning - uses, the rule based approach. [HG14] [QHZ+10] VADER is a very widely used model in the field of sentiment analysis, being notably popular for usage in social media post analysis [EY19]for its versatility with slang, acronyms, and other internet texting patterns. Judging from a few examples, VADER did very well in providing appropriate scores for the negative, positive and neutral tones from the tweets. Extending this, it is helpful for sentiment analysis on Reddit as well, especially because the length of each post is much longer and the number posts I collected was also larger.

Topic modeling is a branch of text mining that involves processing large corpora to produce a set of meaningful topics discussed in the text. [MB13] The most popular methods of topic modeling are logistic normal distributions, bayesian theorems, LDA, and matrix factorisation. [VK20] The method that I employed is HDBSCAN through BERTopic. HDBSCAN is an unsupervised clustering algorithm [MHA+17] that has had comparative results to other methods of topic modeling including. For example, in a study conducted by Egger and Yu finds that BERTopic produces topics just as well as other methods, but is slightly more specific. [EY22] This is great for social media analysis since many people talk about many nuanced topics in the war, topics that another model may classify as being similar or possibly the same.

4 Methodology

4.1 Data collection

Possible projects: the Russian/Ukraine conflict. We downloaded the reddit comment and submission dumps present on the academictorrent site from March-June 2024 in the form of zst files. These file dumps were made via the Pushshift database [BZK+20], a database that returns a copy of the reddit data requested. This data is constantly scraped from the Reddit site and stored on the Pushshift database, keeping it updated. There are 2 zsts for each month with one being the comments and the other being the submissions. This means 8 zsts in total, which were then read through line by line using a python script to extract the jsons of the posts from the required subreddits. These jsons were put, in no particular order, into 2 ndjson files - one for the submissions and the other for the comments. Every json also contains a key for the timestamp of the post or comment which make time series plots simple. Along with this data, we downloaded the dataset Zhu et al compiled from the github they put the data on. It also includes data on Russia's initial occupation of Crimea from reddit in 2014 which I ignored. The data collected is first separated by comments and submission, then by date, and after that even by subreddit. The data is in csv format, one notable characteristic of this data is that it is not as comprehensive for each post as the data I collected is, due to Pushshift adding more categories for each post since 2022. Each submission from the dataset has 100 different data points while the posts that I collected have 110. The number of data points for a comment from the my dataset is 71 while it is 53 for a comment from the researchers' dataset.

Subreddit	**Submissions**	**Comments**	**Active Users**	**Avg Replies/Post**	**Avg Score/Post**
ukraine	12069	263538	33196	16.03	450.57
ukraina	876	8073	2351	7.36	37.82
RussiaUkraineWar2022	1282	22093	6986	6.46	146.00
UkrainianConflict	15830	288345	27631	13.15	179.21
UkraineConflict	2677	19003	3640	6.22	44.92
UkraineWarVideoReport	11899	526694	64568	34.69	508.22
UkraineWarReports	206	1400	665	5.92	35.59
UkraineInvasionVideos	2636	13572	3869	4.27	55.91
war	1625	14384	5373	3.89	11.66
CombatFootage	8115	340352	54235	30.65	402.90
CredibleDefense	314	40280	2353	115.31	29.83
geopolitics	3893	107311	14576	23.04	38.30

Table 1: Reddit communities collected data.

Category	**Subreddit Name**
R-U Conflict	*ukraine, ukraina, UkraineConflict, UkraineWarVideoReport, RussiaUkraineWar2022, UkrainianConflict, UkraineWarReports, UkraineInvasionVideos*
Military Related	*war, CombatFootage, CredibleDefense, geopolitics*

Table 2: Reddit communities we focus on.

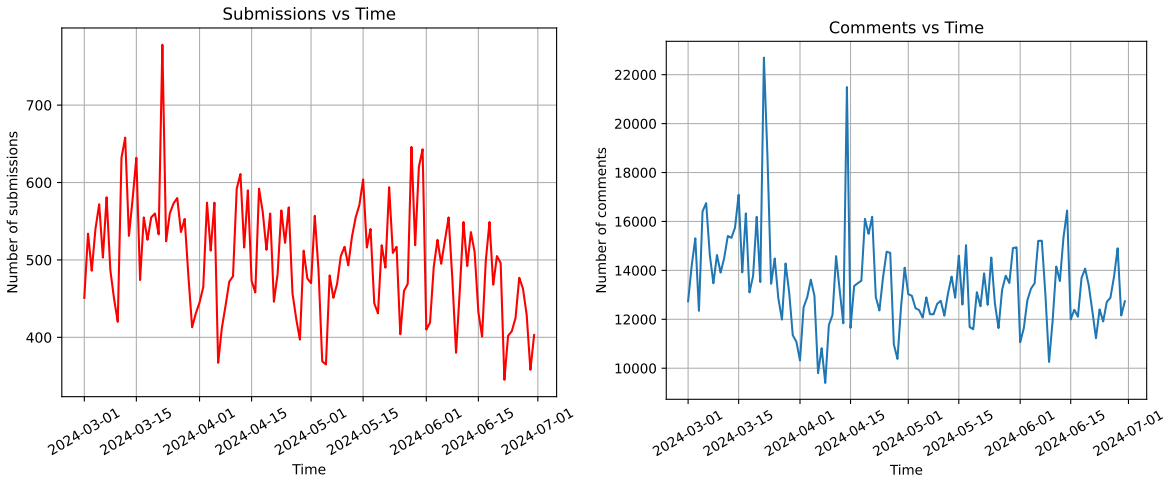
4.2 Methodologies used

How the BERTopic analysis was done was using the title and content of each post, the embedding model used was the paraphrase-MiniLM-L6-v2 for its relatively lighter computation compared to the other BERTopic models. [Rei19] Using this model and the data I collected, I generated the top 40 most popular topics in the dataset along with their top 5 words including bigrams and trigrams. I also plotted a histogram plot using sns of the frequency of a post having a certain number of words in the title as well as the number of words in its selftext. For the researchers data, I did not produce trigrams simply because of computation time: the clustering process as well as the dimensionality fitting took

a very long time to process the 61422 posts. The BERTopic analysis was done for the whole dataset at once rather than splitting it to gain a holistic view on the topics talked about

A separate, singular BERTopic was done for the comments, giving me the most popular topics in the whole 4 month period of my data.

This study utilises VADERNET for the sentiment analysis, largely due to ease of use and accuracy. The sentiment analysis used a sentence by sentence approach. Due to this analysis not utilising stance detection, any sentiment detected in a sentence will be unknown as to whether or not it is towards Russia or towards Ukraine. Therefore, the posts were separated into sentences and the sentences were further subdivided into 3 classes: sentences containing words pointing towards Ukraine, sentences containing "Russia", and finally sentences containing both. The sentences with both were simply ignored in the calculation of sentiment for the the lack of stance detection use in this paper [KC20]. The "Ukraine" and "Russia" classes were then gone through to calculate sentiment for, sentence by sentence, to then calculate the mean, maximum and minimum for each day, plotting its time series. All of the calculations and graphs were done using the compound score metric that VADER calculates.



(a) Submissions

(b) Comments

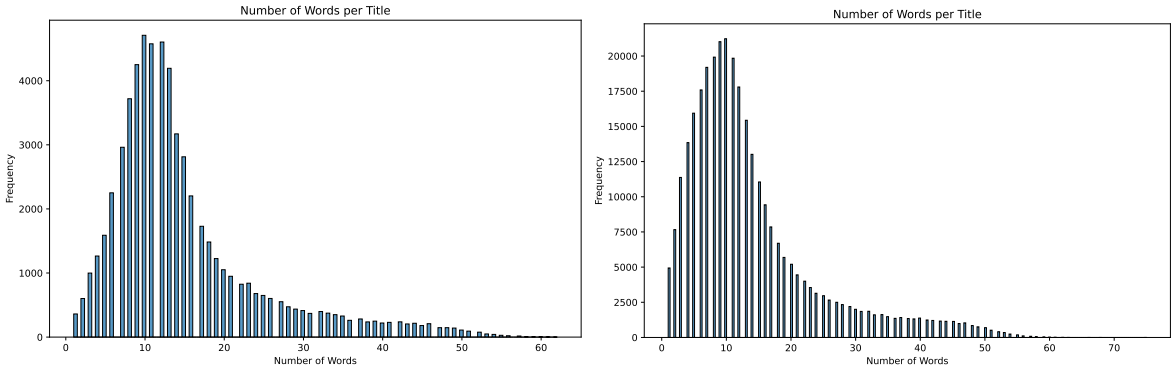
Figure 1: Number of submissions and comments per day

5 Findings

5.1 Longitudinal Analysis

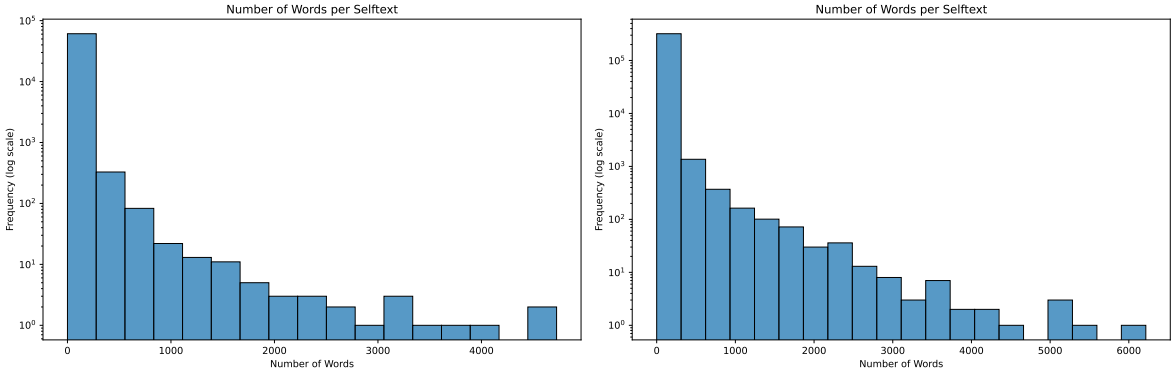
Here put the plots that you did on posts per day and so on and analyse the peaks in activity. What events took place when we have peaks in activity, and so on.

In Figure 1 we plot the number of submissions per day in the subreddits we analyse. The day with the highest number of submissions was March 22, 2024, lining up perfectly with one of Russia's largest missile and drone attacks on Ukraine. At daybreak on this day, Russia deployed 88 missiles and 63 drones to several major locations in Ukraine including Kharkhiv, Zaporizhzhia, and Lviv [Kar24].



(a) Collected dataset (b) Researcher dataset

Figure 2: Frequency of posts against title length for both datasets



(a) Collected dataset (b) Researcher dataset

Figure 3: Frequency of posts against selftext length for both datasets

5.2 Topic Analysis

The total number of active users on my dataset including all the studied subreddits is 163967. Compared to the previous 546593, the current number is 30% of what it once was.

In figure 2, we can see the distribution of the number of posts containing a certain number of words in the title on both my dataset and the researchers' dataset. In order to normalise the shape for both datasets, the y axis was set on a log scale. The frequency is higher for all the values due to its size but the shapes are slightly different. The collected dataset has a smaller number of posts with a shorter title in proportion compared to the researchers' dataset. Otherwise, the shape is exactly the same with the peak for both graphs being around 10-11 words.

Similar to this graph, the graphs presented in figure 3 show the distribution of posts against the length of their selftext. Again, the y axis was set on a log scale to normalise the shapes and we can see that they are same until the right end of the graphs. For the researchers' dataset, there are a number of posts have crossed the 5000 and even 6000 word mark, while on the collected dataset, there are not any posts with selftexts greater than 5000 words.

Though topic 0 in BERTopic is labelled as a topic that has various words that, although similar, are not correlated enough to be labelled under a topic, the topic 0 from my BERTopic analysis appears to be a valid topic discussing Putin. The difference between tones in the topics discussing Putin in both datasets is notable. The top five phrases in the researchers dataset were "Putin none", "none

Putin", "fuck Putin", and "none fuck" while the top five phrases in my dataset for Putin was "Putin", "Putin is", "Vladmir", "Vladmir Putin", and "he". This difference in tone was likely caused by the initial outrage and thoughts regarding the invasion of Ukraine. One issue with the topic modeling here is that it does not recognize variations of what users call Putin like "Putler" - a play on words comparing him to Hitler which many posts refer to him as nowadays.

Author	**Subreddit**	**Selftext or Title**
MS_09_Dom	geopolitics	*It's been speculated for a while that the US-Israel alliance is approaching an inflection point, namely in how younger generations are much more sympathetic to Palestine compared to previous ones*
Beelzebubs-Barrister	CombatFootage	*Israel bombs civilians recovering food air-drops*
BennamStyle	geopolitics	*Considering that South Africa are declaring that what Israel is doing to Palestine is genocide, why aren't they saying the same about China and the situation with the Uyghurs?*

Table 3: Example posts discussing the Israel-Palestine situation

Due to the conflict in Palestine being a more topical area of interest, the focus of many of the subreddits I worked on was shared with Ukraine and Palestine. The number 1 topic by BERTopic reflects this with it being the Israel Palestine conflict, its top words being "Israel", "Gaza", "Palestine", "Hamas", "Palestenian". Table 3 refers to examples of some posts discussing this topic. This topic is barely discussed in the researchers' dataset despite it being active at that point, likely due to it increasing in popularity once the bombing started in Hamas and more recently Rafah.

This discussion is mostly carried out in the subreddits that are unspecific to the Russo-Ukrainian war like "war", "geopolitics" and "CredibleDefense". In particular, they discuss this topic in much detail on the geopolitics subreddit. In the 3rd example post, the poster brought up 4 different countries to involve in discussion about 2 conflicts separate to the Russo-Ukrainian war.

The second most popular topic is one that discusses people of power. The word "removed" is the most popular word in this topic, followed by the phrases "removed removed" and "removed this". Though the last 2 are due to posts with text that has been modified due to the removal, the first word has a very high usage because users discuss Putin, Zelenskyy, and other people in office, sometimes saying that they should be removed from power, particularly with Putin. Users also discuss their own posts which have been removed from the site and question why they were removed as shown in table 4

The next most common topic discussed on my dataset is one talking about weapon strikes Russia conduct on Ukraine. The 5 most common phrases include "weapons", "to strike" , "targets", "strike", and "Ukraine to". This idea of attacks is one of the most obvious topics of discussion in war and this idea is one that stayed consistent in the researchers' dataset.

One of the main ideas in the rest of the topics refer to many events of the war like power plants

Author	**Subreddit**	**Selftext or Title**
Visual-General-6459	UkraineWarVideoReport	*Open question to the Mods, few times now I've had posts removed for being "reposts" that clearly are not. I've messaged you multiple times to see what was up. Never got a response.*
Legitimate-Champion3	UkraineWarVideoReport	*Pro-Russian traitor removed from office.*
Sniper9Xx	CombatFootage	*So I posted a video of a Hamas soldier sniping 3 idf soldiers but I don't get why the video was removed as this wasn't an attack on civilians but an attack on military targets.*

Table 4: Examples of posts discussing the topic of removal

Author	Subreddit	Selftext or Title
Accomplished-Army-31	UkrainianConflict	*(FOOTAGE) RUSSIAN Helicopter SHOT DOWN By UKRAINE —— Compilation Feb 26*
Arty_beaver	UkraineWarVideoReport	*May 7, Ukraine defenders in the Donbass shot down a Russian Mi-28H attack helicopter and destroyed another half a hundred enemy vehicles.*
Max_Kos_	UkraineWarVideoReport	*Ukrain army group of BMPs being air missiled near Nikolaev, late may 2022. Presumingly with a controlled missile strike from Ka-52 copter.*

Table 5: Examples of posts discussing the topic of helicopters

Author	Subreddit	Selftext or Title
vancouver_reader	UkrainianConflict	*As the world sanctions Russia, China takes note*
UkraineWarNewsReport	UkrainianConflict	*This sanctions package includes other hard-hitting measures: de-Swifting the largest Russian bank Sberbank, banning 3 more Russian state-owned broadcasters, and sanctioning individuals responsible for war crimes in #Ukraine*
Moneymaker2106	UkraineInvasionVideos	*EU foreign policy chief Josep Borrell and Ukrainian Foreign Minister Dmytro Kuleba in a phone call on Sunday, May 29, discussed further EU sanctions against Russia.*

Table 6: Examples of posts discussing the topic of sanctions

being attacked, or technology used in the war such as FPV drones and tanks. Another prominent idea in these topics is the political discussion regarding the war. They often discuss Russian propoganda, NATO, the media, and aid from governments.

Overall, the topic discussion of the data researchers collected and studied is more different to my data than expected. The most popularly discussed topic in this dataset is about Russian attacks or scouting involving helicopters, similar to the theme of war technology discussed in my dataset. The most popular phrases are "helicopter", "russian helicopter", "helicopters", "mi", and "ka". "mi" and "ka" actually refer to models of Russian helicopters: Mi-28 and Ka-52, which are attack helicopters that have been used heavily during the war, even now [Nik24] and in table 5 examples of posts discussing this topic can be seen. This topic is mostly discussed in the subreddits involving video footage about the war like UkraineWarVideoReport and CombatFootage.

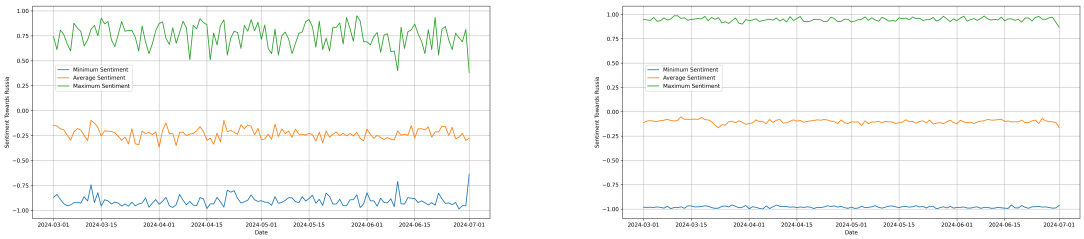
The second most popular topic on the researchers' dataset describes economic sanctions. The most popular phrases are "sanctions", "sanctions on", "sanctions against", and "of sanctions". They often discuss this topic with regards to countries that have stopped trading with Russia or Ukraine but more often than not it is Russia. As can be seen in table 6, various different posts are discussing sanctions placed on Russia and potential further sanctions that may be placed on Russia as well.

The third most popular topic discussed is about Ukrainian refugees. 5 of the most popular phrases include "refugees", "ukrainian refugees", "refugee", "refugees in", and "asylum". Due to bombing in major cities like Kyiv and Kharkhiv, many civilians were caught in the crossfire and left to neighbouring countries like Hungary, Poland, Moldova etc. On top of fleeing to neighbouring countries, many countries part of the EU offered asylum to the Ukrainian refugees like Germany [Rau24]

A large difference between my dataset and the researchers' dataset is the amount of Russian in the posts. Including topic 0, there are 3 topics that are in Russian in the researchers' dataset but in my dataset there is only 1. Only a small number of people use Reddit in Russia but the extent of discussion of the war seems to have fallen on the studied subreddits for Russian speaking users as well.

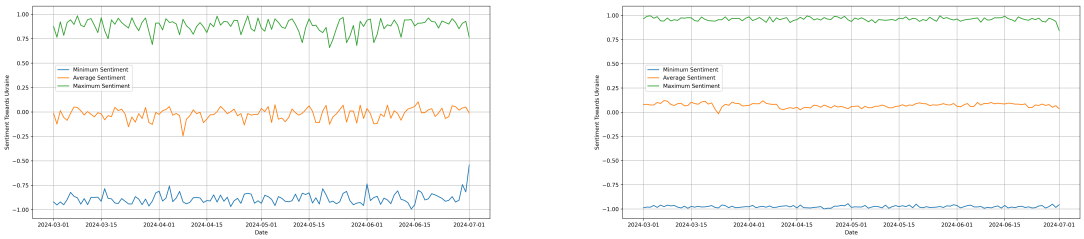
Other topics discussed in the researchers' dataset include other countries, weapons, people in power, and sites of war or bombing, mainly power plants, nuclear reactors and other radioactive areas.

Using BERTopic, which employs HBDSCAN, we are able to plot a hierarchical topic graph which encapsulates the sub topics into sections that can be grouped up into more general topics. Through this, an overview of the discussions taking place can be understood for both sides.



(a) Submissions (b) Comments

Figure 4: Sentiment of submissions and comments per day towards Russia for my dataset



(a) Submissions (b) Comments

Figure 5: Sentiment of submissions and comments per day towards Ukraine for my dataset

Full BERTopic results can be found in this github repository

5.3 Sentiment Analysis

In figure 4 we plot the sentiment towards Russia against time for the comments and submissions of my data. The average sentiment of the comments and the submissions of the data I collected also reflect the attack on the 22nd of March 2024 with both reaching a minimum on the day after the attack at -0.1642 and -0.3335 respectively, where the minimum for comments was a global minimum. Due to having a smaller number of submissions compared to the number of comments, the average of the submission sentiments was less consistent and as a result the sentiment is more erratic and at times much more negative than the comments'.

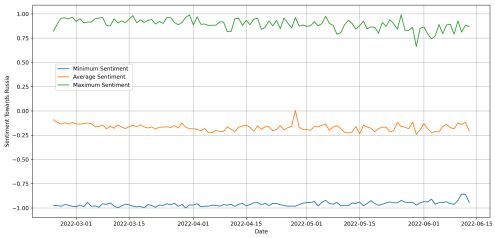
The day with the lowest average sentiment for the submissions was the 1st of April 2024 at -0.3684, this although the sentiment score average is oddly low here, it is likely due to the Red cross' humanitarian aid team being unable to reach Kyiv, Russian soldiers facing radiation exposure from Chernobyl, and an oil Depot being struck in Russia.

One important feature of the submission sentiment towards Russia for my dataset is that the average sentiment seems to deviate around -0.1 to 0 and in 5 the average sentiment towards Ukraine for my data also deviates around 0 to 0.1. A huge factor behind the submissions being more negative for both countries is likely due to Reddit being a social media platform - they often have misinformation,

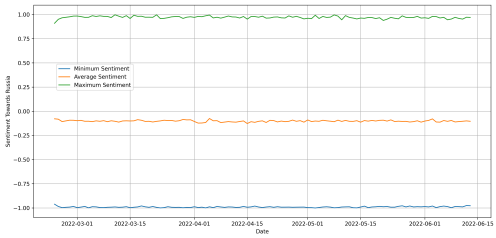
bias, and a lot of negativity as well compared to positivity. Regardless, the sentiment towards Ukraine is more positive than the sentiment for Russia

The sentiment analysis of the researchers' data in figure 6 reveals that Russia was viewed in roughly the same light as it is viewed now. The graph of both the comments and submissions still deviate around a value of -0.1 to -0.2. The overall sentiment in the comments appears to be slightly lower than the sentiment towards Russia for the comments in my dataset, again attributing the outrage associated with the beginning of the war.

However, the difference is more clear when viewing the sentiment towards Ukraine. The average sentiment towards Ukraine was less positive than it is now with it deviating around 0.05 to 0 for the comments and 0 to -0.05 for the submissions. It shows a trend of increasing positivity in sentiment as the days went on, due to sadness and outrage laced in the comments in the initial war days and then positivity and encouraging ideas as the days went on. Overall, the data from my data set appears to be more constant against time with it not clearly displaying any trend of decreasing or increasing.

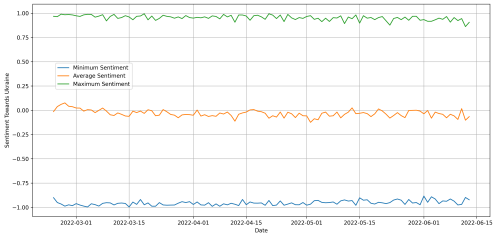


(a) Submissions

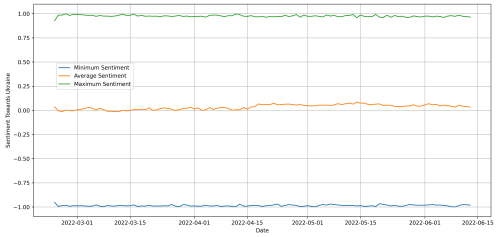


(b) Comments

Figure 6: Sentiment of submissions and comments per day towards Russia for the researchers' dataset



(a) Submissions



(b) Comments

Figure 7: Sentiment of submissions and comments per day towards Ukraine for the researchers' dataset

6 Discussion

Although the results were good and many nuances of the war were found and discussed, there are some flaws in the methodology. Firstly, the fact that I did not use stance detection means although there might have been a sentence containing Ukraine and not Russia, it may not necessarily have been referring to Ukraine in some of the comments made in the post. Therefore, there likely exist a number of incorrectly identified sentiments in the data, though taking an average every day likely helped in suppressing this issue. In addition to this many posts may have included sarcasm, irony or had more context which the VADER model would not have captured, leading to erroneous calculations for sentiment. Furthermore, the negative posts attached to a country may not mean that the poster views the country negatively, but that they may be sad about the country's situation and are concerned for it, again causing more sentiment identification errors.

Secondly, the fact that this methodology only considers text data means the subreddits like CombatFootage or UkraineWarVideoReport sacrifice huge amounts of their data. As a result, even though these subreddits have a lot of posts, the actual weightage of the subreddits in the analysis is far less, and this is worsened by the fact that video footage has become a huge part of peoples' opinions on any war that takes place. Furthermore, the data from the subreddits studied, although mostly English, contain many posts in languages like Russian but VADER sentiment works best with just English, further reducing the precision of the sentiments calculated.

Thirdly, there is always a large amount of bias involved in any social media dataset, thus it is difficult to draw concrete conclusions from the data. The data is limited only to places where Reddit is popular so the data will likely be biased to the peoples' opinions in certain countries like USA, UK, or India [Wor24]. Even with the use of moderators, not every post is moderated or moderated properly, therefore many of the posts on both my dataset and the researchers' dataset include incorrect information and baseless conjectures. Another huge issue with this type of data is the misinformation posters, especially bots that purposefully post information that they know to be false at mass on many different communities. These posters make it difficult to identify topics being discussed and topics simply being spammed in the different subreddits.

Another issue with the topical analysis is that I chose not to calculate probabilities so I could not offer the confidence level of each topic calculated by the BERTopic analysis. As a result, it is difficult to understand how pervasive each topic was in the period of discussion.

7 Conclusion

As for the first research question about how sentiment towards Russia and Ukraine changed over time, the results suggest that it has not for Russia but that it has for Ukraine in the fact that it is now more positive than before. This therefore means that my hypothesis that sentiment towards Russia would have decreased is then incorrect and that it has actually stayed more or less constant. My other hypothesis about the huge disengagement in the war is abundantly clear from the data. Both span a similar time period of 4 months yet my dataset had around 30% of the users compared to the researchers which is overall predictable as other events occupy the media's attention but considering that this is a war that is still going on the drop in engagement is quite significant.

For the second research question about how the topics discussed have changed over time, the results say that the users discuss a variety of topics related to the war like attacks, political figures, countries with conflicts, and events or soldiers in the war. These topics have changed slightly over time with more emphasis being placed on attacks and conflicts nowadays. It was slightly unexpected that the most popular topic in my dataset was about the Israel-Palestine conflict and that the political focus had shifted more to the war perspective. This does however make sense as people are likely less concerned over politics as the war goes on for an increasing number of videos and posts describe the war in detail.

Overall, my hypothesis that the interest in the war has fallen is definitely true and clearly corroborates my hypothesis of a dwindling interest. Though the researchers' dataset was collected at the time when the war started, making it a very popular area of conversation at the time. Even so, it is evident that interest in the war has fallen.

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